YZM5110 - Research Project

Course weight: 30%

Format: Individual

Due: Week 7, 18 November 2026, before the presentation session

1. Purpose

In this project, you will study a topic in multimodal learning closely enough to explain how its methods work, where they fall short, and what might improve them. You will bring together a detailed reading of the literature and an experiment of your own.

Later on, you will choose a study involving at least two modalities, reproduce it, and then implement and evaluate an improvement to one aspect of it.

A limitation you notice while reading should help you decide what to test. Reproducing the method gives you a baseline, and your next experiment should show whether your change makes a difference. You need runnable code, your own measurements, and a careful explanation of what they mean.

2. Select and evaluate the literature

Begin with a question you can reasonably investigate. Find the papers that introduced the main ideas, recent work that builds on them, and competing approaches to the same problem. Explain how you found and selected these papers. Your selection should give the reader a thorough understanding of the question and show that you have examined the central papers in depth.

Throughout your report, make it clear when you are describing the authors' findings, offering your own interpretation, or reporting a result you measured.

You are expected to analyze at least 5 papers.

Prepare a comparison matrix with one row per paper and these fields:

Field	What to evaluate
Question and contribution	What problem is addressed, and what is actually new?
Modalities and assumptions	What information is available during training and deployment?
Mechanism and objective	How does information interact, and what signal trains the model?
Data and protocol	Which datasets, preprocessing, split units, and evaluation conditions are used?
Baselines and evidence	Are comparisons fair, and do ablations support the explanation?
Resources and reproducibility	What hardware, code, checkpoints, and configuration details are available?
Limitations and connections	Where does the method fail, and how does it relate to the other papers?

3. Reproduce a concept or result

Choose a specific paper with a claim that you can test. Explain what you expect to observe and what result would lead you to question the claim.

Implement the concept yourself or adapt an existing, attributed implementation that you understand. Run the experiment, record the source revision, dependencies, checkpoint, data version, preprocessing, seeds, configuration, and execution commands.

You may need to use less data, a smaller backbone, or a lower compute budget than the original study. That is a reasonable way to keep the project feasible, but it changes what you can conclude. Describe each difference from the original setup and explain which comparisons still hold. Keep the paper's published scores clearly separate from the results you obtained.

Show the original reported result and your measured baseline separately. Investigate discrepancies before attributing them to the method. Explain the limitation and keep claims appropriately narrow.

Make sure you record this process in your report.

4. Improve one aspect

Once you understand the baseline, choose one weakness to address. You might try to improve alignment, reduce resource use, handle a missing input, or correct a recurring failure. Explain why you expect the change to help before examining its results. This gives your experiment a clear question to answer.

Implement the change and evaluate it against your reproduced baseline under matched conditions. Include a control or ablation that helps isolate the effect of your intervention. Report task quality and an appropriate resource measure, such as latency, peak memory, or training time.

For example, you might investigate whether a small gating change reduces failure when one sensor becomes unreliable. Compare against the original fusion method, test clean and corrupted conditions, and check whether robustness comes at the cost of clean-input performance.

Your proposed change may not improve the result. If that happens, examine why and explain what you learned. A carefully conducted experiment is still valuable when its outcome is negative. Report unfavorable examples as well as favorable ones, and do not adjust the method using the test set.

Make sure you record this process in your report and the results are available in your final repository.

5. Deliverables

Detailed report

Submit your report as `<LastName>_YZM5110_Research.pdf` and put the repository URL on the first page.

Use this structure:

1. Research question, scope, motivation, and contributions.
2. Literature search approach and critical synthesis.
3. Technical explanation of the selected concept or method.

4. Reproduction target, implementation, and deviations from the source.
5. Proposed improvement and its rationale.
6. Experimental design, datasets, splits, baselines, metrics, and resources.
7. Results, controls, uncertainty, and failure analysis.
8. Limitations, conclusions, and unresolved questions.

Runnable repository and results

Provide a public repository which includes setup instructions, dependency versions, licensed data-access instructions, exact commands, configurations, saved outputs, and scripts that produce the reported tables or figures. Pin the submitted version with a commit or tag.

Do not upload credentials, restricted datasets, or large checkpoints unnecessarily. If a full run is expensive, provide a small test alongside instructions for the reported experiment. State which results require additional resources.

6. Assessment criteria

This project is worth 30% of your course grade. When reviewing it, I will consider the following aspects together.

Criterion	Evidence of strong work
Literature depth and synthesis	Detailed, accurate comparisons and a justified research gap
Technical understanding	Clear explanation of mechanisms, assumptions, and objectives
Reproduction	Runnable experiment, traceable results, and transparent deviations
Targeted improvement	A reasoned intervention evaluated against an appropriate baseline
Experimental rigor	Fair controls, valid splits, honest uncertainty, and failure analysis
Reproducibility and communication	Usable repository, detailed report, and a defensible presentation

I will look at how you reached your conclusions, not only at whether your final score improved. Careful reasoning, fair experiments, and an honest account of limitations matter throughout the project.

7. Integrity and AI-use policy

You are welcome to discuss papers and exchange general feedback with classmates. The submitted work must still be your own. Do not divide the project between students or share project-specific code, experiment outputs, analysis, reports, or slides.

Disclosed AI assistance for search, coding, or editing is permitted for this research project. You remain responsible for reading the sources, understanding the implementation, designing and running the experiments, checking citations, verifying every claim, and defending the conclusions. Identify the tools used and the specific contributions they made.